OSCAR DE LA CERDA

Portfolio: oadlc.github.io
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EDUCATION

Yale University, New Haven, CT — May 2028
- ABET-accredited B.S in Electrical Engineering, Certificate in Programming | GPA 3.83
- Relevant Coursework*:* Circuits and Systems Design, Electromagnetic Waves and Devices, Introduction to Computer Engineering, Introduction to Communications and Control, Data Structures and Programming, Digital Systems

University of Iceland, Reykjavik, Iceland — June – July 2025
- Relevant Coursework*:* Renewable Energy Technology, Politics and Economics of Renewables

EXPERIENCE

L'SPACE Mission Concept Academy (NASA-affiliated), CDH Engineer — September 2026 – Present
- Designing the command and data handling subsystem for a robotic lunar payload intended to operate in the ice-bearing polar regions, working with an interdisciplinary team of 17 that follows NASA proposal review and procedures
- Drafted the majority (~11) of the CDH subsystem requirements using SMART criteria, with traceability to parent requirements, rationale, and verification methods
- Authored the CDH section of the System Requirements Review, covering the flight computer, storage, telecommunications, and software architecture; a subassembly budget for mass, power, and TRL; a command and data flow diagram; and risk and mitigations for link loss, processor or storage failure, and data corruption
- Ran a communications trade study comparing fixed, gimbaled, and phased-array antennas on link robustness, signal quality, power, and mass

Quantum Network Research, Undergraduate Researcher — June 2026 – August 2026
- Automated fast-steering-mirror (FSM) sweep sequences across a quad-cell photodetector sensor in Python, mapping boundary conditions and characterizing sensor response across the full field of view
- Designed and implemented a closed-loop control system to stabilize laser pointing on a quad-cell sensor under simulated FSM disturbances, applying circuit and system-level analysis to inform controller design

Yale Undergraduate Aerospace Association, Avionics Sub team member — November 2024 – May 2026
- Supported instrumentation and testing for flight readiness: designed and simulated an ADC-based data acquisition circuit enabling a Teensy microcontroller to collect pressure, voltage, and current sensor data during pre-fuel testing
- Developed a complete avionics sensor-suite schematic integrating GPS, IMU, altimeter, MCU, and communication modules, supporting hardware integration and flight data acquisition on a rocket vehicle platform

PROJECTS

Two-Axis Ball-Balancing Stabilizer Platform, Independent Project — June – July 2026
- Designed a 3RPS-style two axis stabilizer around a custom ESP32-WROVER-CAM PCB carrier board (schematic capture and layout in KiCad) driving three MG966R servos – end-to-end hardware, firmware, and controls integration
- Debugged firmware level GPIO/PSRAM and PWM channel conflicts during board bring-up, and migrated servo control from HTTP to UDP to eliminate control-loop latency
- Built a Python/OpenCV vision pipeline with HSV ball tracking feeding a dual-axis PID controller driven by a law of cosines inverse-kinematics solver

SKILLS

- Test & Debug: oscilloscopes, multimeters, board and system-level debugging, instrumentation & data acquisition, hands-on lab test experience, , breadboard prototyping, soldering
- PCB & Avionics: schematic capture, PCB layout, assembly & manufacturing (own PCBAs), board bring-up, analog/digital circuit design, ADC circuits, power systems, sensor integration (GPS, IMU, altimeter), motors
- Tools & Languages: Altium, KiCad, LTspice, Zemax, C, C++, Verilog, MATLAB, Python, Racket, PowerShell